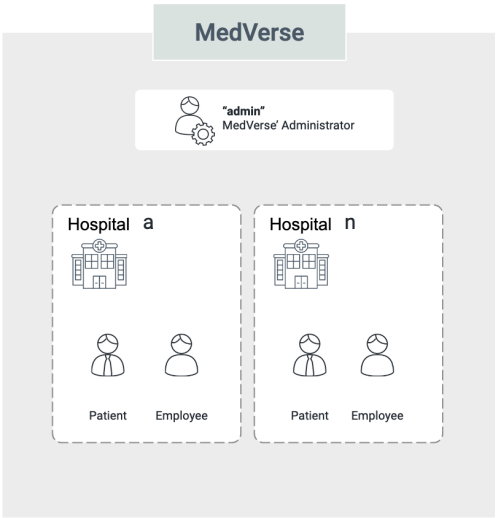
B2B CIAM

If you have a business that offers Business-to-Business (B2B) solutions, you need to define the structure of your organization in WSO2 Identity Server so as to represent all your partner/supplier organizations. You can then share your applications and services with your partner/supplier organizations and allow them to manage their own identity and access management requirements.

- All partner/supplier organizations of your business should be set up as organizations of your organization (root) in WSO2 Identity Server.
- Once the organizations are set up, you should onboard administrators to them. These Administrators can then use a separate administration portal created using WSO2 Identity Server's B2B APIs to manage their respective organizations.
- The organization (root) needs to share applications with its organizations so that the users managed by the organizations can log in and use them.
- If the partner/supplier organizations of your business have a requirement for their own custom applications, they should be able to set up applications in their organizations itself.



Let's use the use case in the diagram to demonstrate how we can configure Organization, Users and Applications to cater a B2B use case. As you can see, **admin**

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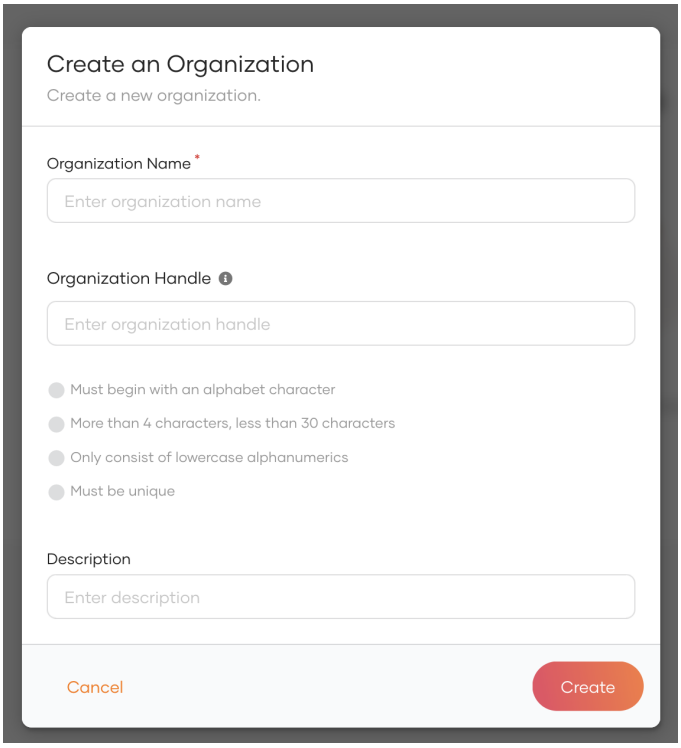
is the super Administrator of MedVerse. MedVerse has two Hospitals registered Hospital a and Hospital n. Each hospital has patients and employees registered separately.

Medverse operates a **Shared Patient Portal** that both **Hospital a** and **Hospital n** need to access. On top of that, **Hospital n** has built its own custom applications: a mobile app for patients and an internal app for staff.

Additionally, Hospital a uses MFA for the authentication of patients whereas Hospital n uses only basic authentication. This can be configured organization wise. Let's see how this can be done.

Onboarding Organizations

1. Login to the WSO2 Identity Server **Console,** using your admin credentials (e.g. admin:admin).
2. In the WSO2 Identity Server **Console,** from the menu click **Organizations.**
3. Then click **+ New Organization**.



4. Enter **Hospital a** as the organization name and click **Create**.

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5. Do the same for **Hospital n**.

Now you have registered Hospital a and Hospital n as organizations in MedVerse.



Create Application and Share with Organizations

Now you need to create the patient portal from the MedVerse root organization, and share it with Hospital a and Hospital n.

In the WSO2 Identity Server **Console** of **MedVerse** root organization**:**

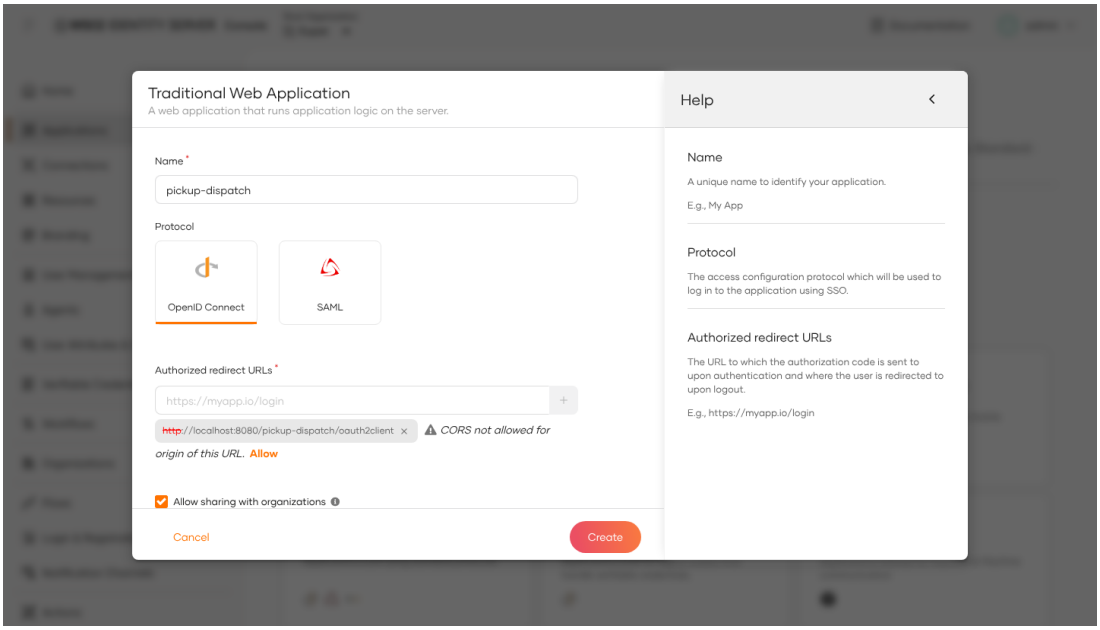
1. Click **Applications.**
2. Click **New Application.**
3. From the given set of templates select **Traditional Web Application** template.
4. Fill the fields in the create application wizard.

    Name: *pickup-dispatch*

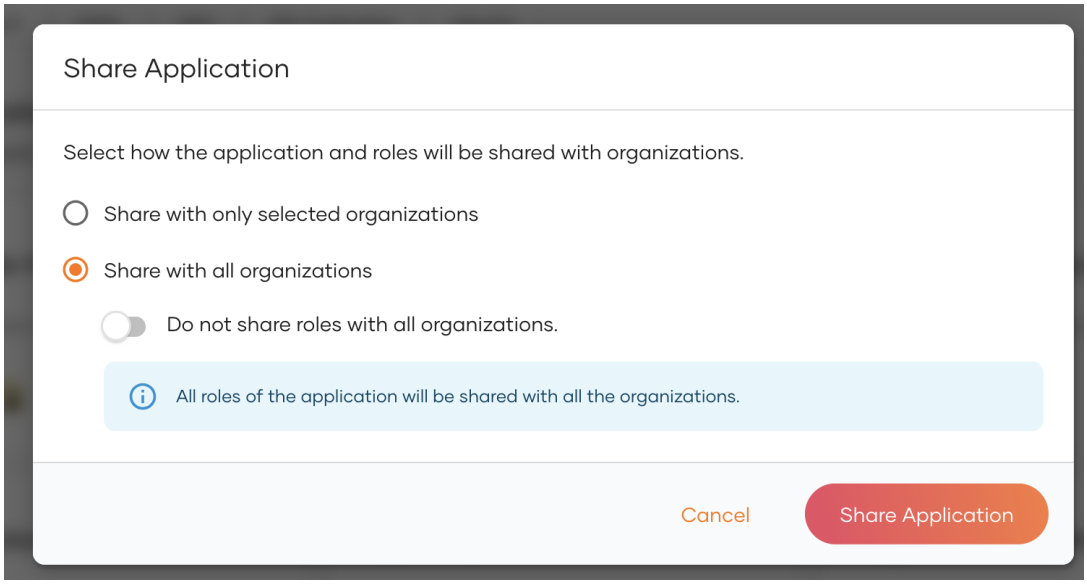
    Protocol: *OpenID Connect*

    Authorized redirect URLs: *http://localhost:8080/pickup-dispatch/oauth2client*

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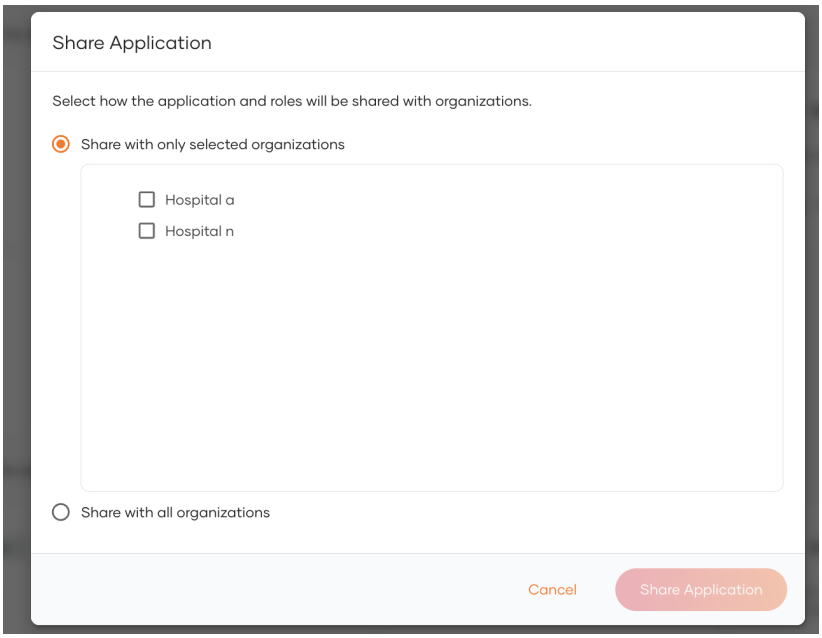


5. Click on Allow sharing with organizations and click create.



6. Select **Share with only selected organizations**.

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7. Select **Hospital a** and **Hospital n** as options and click **Share Application**.
8. Goto Protocol tab and note **Client ID** and **Client Secret**.
9. Open `<TOMCAT_HOME>/webapps/pickup-dispatch/WEB-INF/classes/dispatch.properties` (deploy `pickup-dispatch.war` to Tomcat first, as in Lab 05).
10. Replace **consumerKey** and **consumerSecret** values with **Client ID** and **Client Secret** taken from the newly created application.
11. Save the file. Restart Tomcat: in `<TOMCAT_HOME>/bin` run `catalina.bat stop` and then `catalina.bat start` on Windows, or `sh catalina.sh stop` and then `sh catalina.sh start` on Linux and macOS.

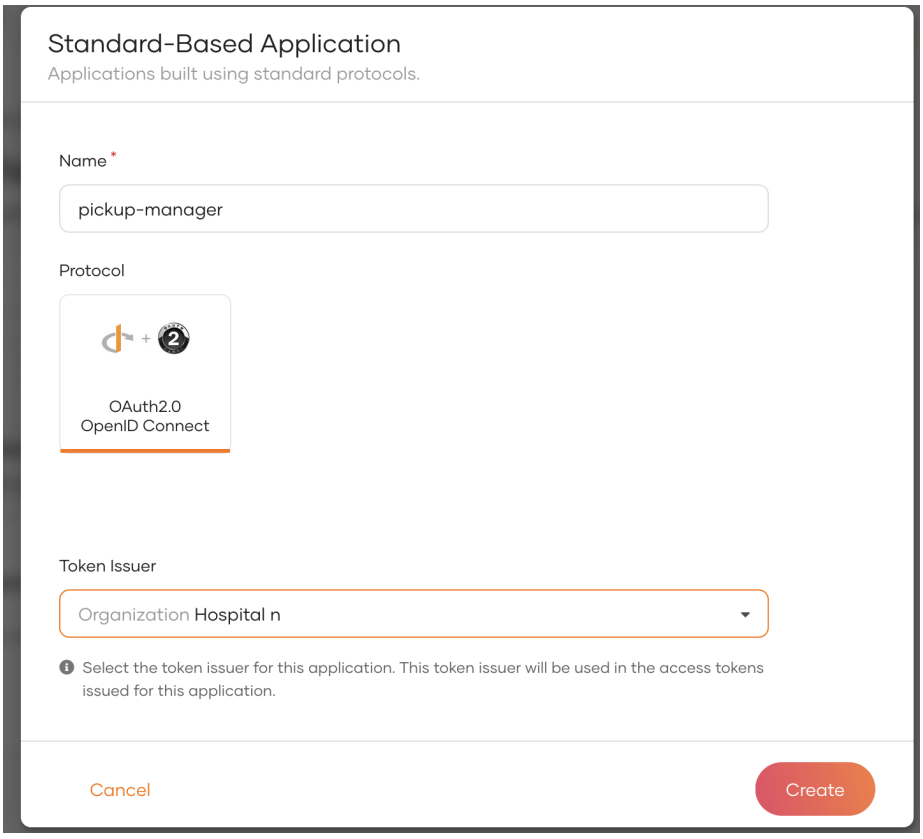
Create Applications in the organization level

Hospital n needs to have a mobile app for patients and an internal staff. Therefore sub organization admin will be creating mobile apps at the sub organization level.

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First, switch to the Console of **Hospital n**: from the menu click **Organizations** and click the switch icon (⇄) next to **Hospital n** (you can also pick **Hospital n** from the organization switcher at the top of the Console). Then, in the WSO2 Identity Server **Console** of **Hospital n**:

1. Click **Applications.**
2. Click **New Application.**
3. From the given set of templates select **Standard-Based Application** template.
4. Fill the fields in the create application wizard.
    a. **Name:** *pickup-manager*
    b. **Token Issuer:** Organization Hospital n



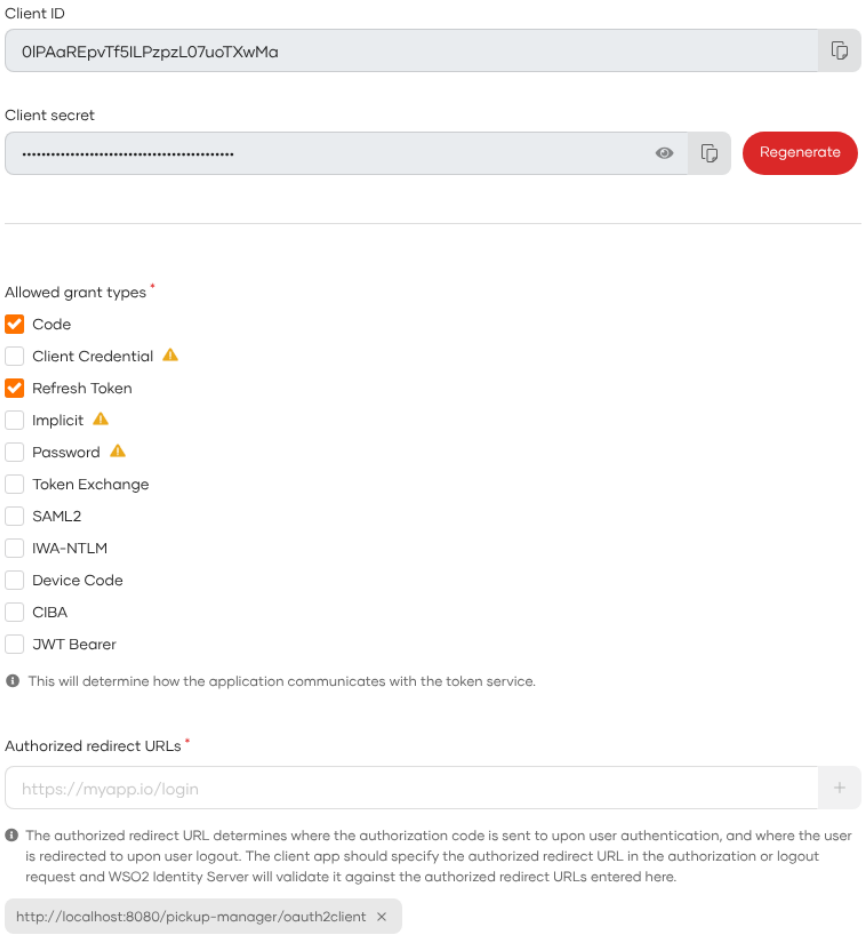
12. Fill the fields in the create application wizard.
13. Goto Protocol tab:

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a. Note **Client ID** and **Client Secret**.

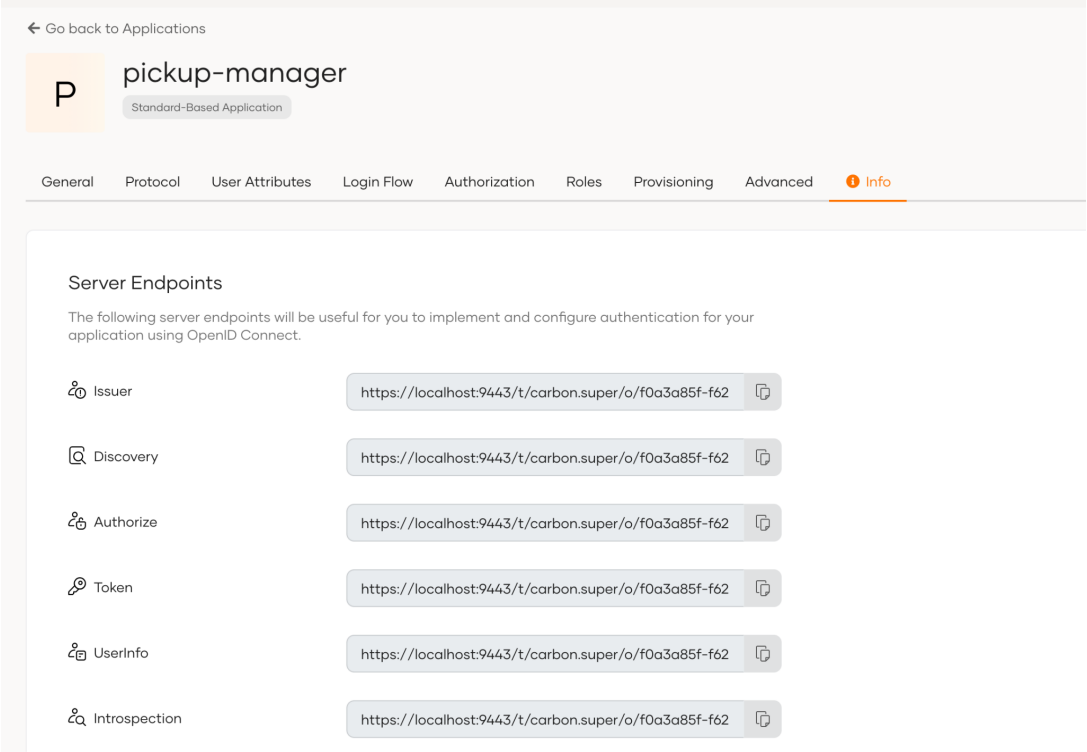
b. Enable **code** under **Allowed grant types.**

c. Configure *http://localhost:8080/pickup-manager/oauth2client as a Authorized redirect URLs*



14. Goto **Info** tab to find the server endpoints needed for application integration.

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15. Open `<TOMCAT_HOME>/webapps/pickup-manager/WEB-INF/classes/manager.properties` (deploy `pickup-manager.war` to Tomcat first, as in Lab 06).
16. Replace **consumerKey** and **consumerSecret** values with **Client ID** and **Client Secret** taken from the newly created application.

```
consumerKey=<Client ID of pickup-manager>
consumerSecret=<Client Secret of pickup-manager>

callBackUrl=http://localhost:8080/pickup-manager/oauth2client
scope=openid email profile internal_application_mgt_view
authzGrantType=code

enableOIDCSessionManagement=false
enableOIDCBackchannelLogout=true
authzEndpoint=https://localhost:9443/t/carbon.super/o/<org-id>/oauth2/authorize
OIDC_LOGOUT_ENDPOINT=https://localhost:9443/t/carbon.super/o/<org-id>/oidc/logout
sessionIFrameEndpoint=https://localhost:9443/t/carbon.super/o/<org-id>/oidc/checksession
tokenEndpoint=https://localhost:9443/t/carbon.super/o/<org-id>/oauth2/token
claimManagementEndpoint=https://localhost:9443/services/ClaimMetadataManagementService

post_logout_redirect_uri=http://localhost:8080/pickup-manager/oauth2client

adminUsername=admin
adminPassword=admin
```

Also update the endpoint properties in manager.properties to the organization endpoints shown in the **Info** tab, as in the example above. Replace <org-id> with the ID of Hospital n:

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authzEndpoint=https://localhost:9443/t/carbon.super/o/<org-id>/oauth2/authorize

tokenEndpoint=https://localhost:9443/t/carbon.super/o/<org-id>/oauth2/token

OIDC_LOGOUT_ENDPOINT=https://localhost:9443/t/carbon.super/o/<org-id>/oidc/logout

sessionIFrameEndpoint=https://localhost:9443/t/carbon.super/o/<org-id>/oidc/checksession

Save the file. Restart Tomcat: in `<TOMCAT_HOME>/bin` run `catalina.bat stop` and then `catalina.bat start` on Windows, or `sh catalina.sh stop` and then `sh catalina.sh start` on Linux and macOS.

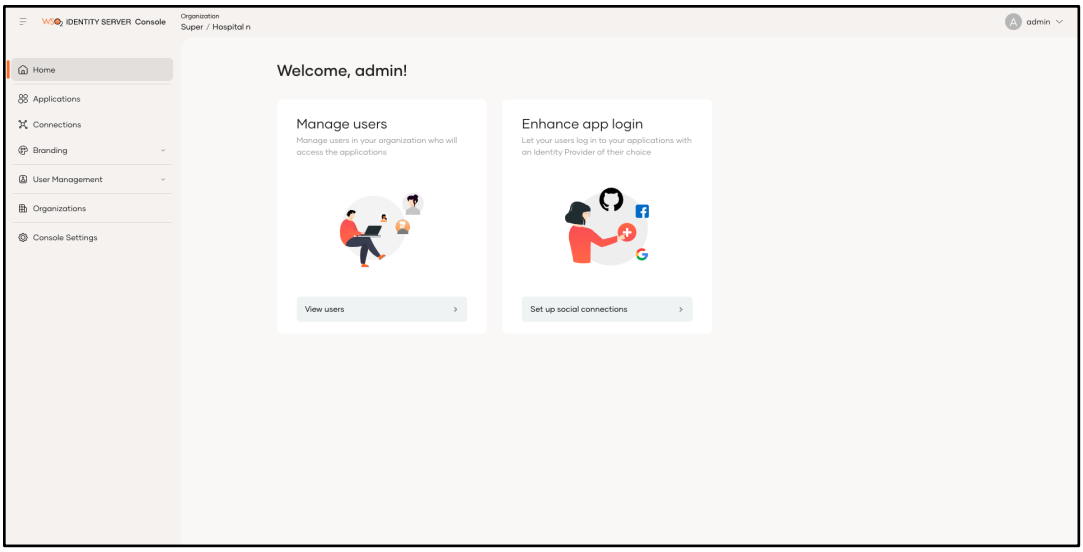
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Onboarding Organizations Users

As of now user **admin** has Administrative privileges over both the hospitals. Therefore admin will be creating users in both **Hospital a** and **Hospital n**.

1. Switch Organization

I. In the WSO2 Identity Server **Console,** from the menu click **Organizations.**
II. In the given list of organizations, click on the switch button of **Hospital n**.
III. You will be redirected to the **Console** of Hospital n.



2. Create Users

I. Expand the **User Management** menu item from the menu and click **Users.**
II. On the **Users page**, click **Add User** and select **Single User** from the dropdown.
III. To create patient **Tommy**, fill all required fields and click **Finish**.

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3. Edit Application

I. Click on the **Applications** from the menu.

II. You will see pickup-dispatch listed as a shared application in the organization **Hospital n**.

III. Click **edit**.

IV. Let's add **Email OTP** as the 2nd step in the authentication.

   A. Click on the plus icon followed by the first screen.

   B. Then click **+ Add Sign In Option**.

   C. Select **Email OTP** as authentication.

V. Click **Update**.

Now follow the same steps from 1-3 for the **Hospital a**.

1. Add a patient name with **Kate**.
2. Leave the Application **Login Flow** as it is.

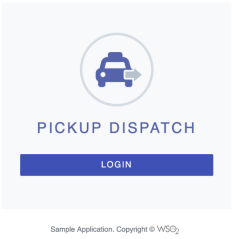
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Try it

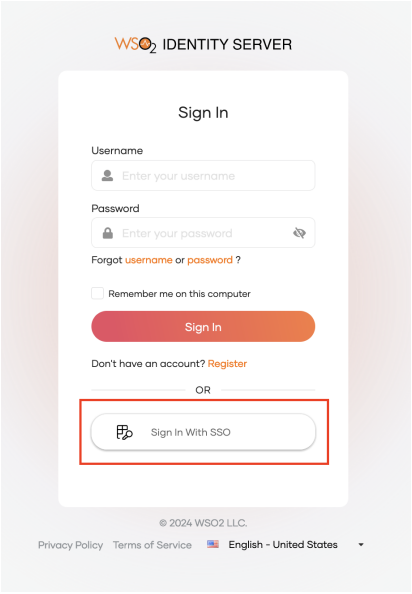
Let's try out the B2B use case using the configured Applications, Organization and Users.

1. Access the pickup-dispatch app via http://localhost:8080/pickup-dispatch



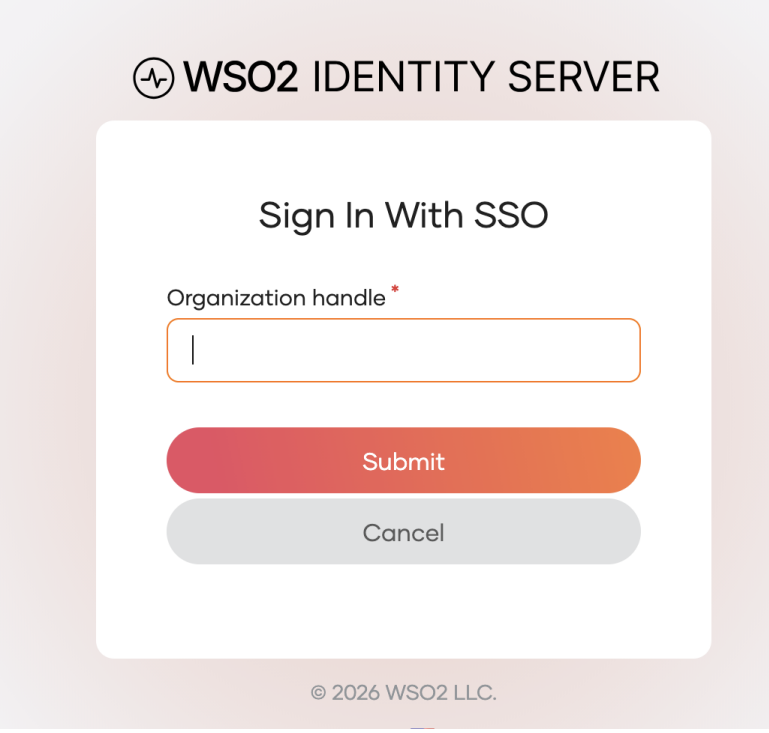


2. Click **Login**. The browser will be redirected to the login page of the root organization
3. Select **Sign in with SSO** option to go to the organization login.



4. Then type the organization handle as **hospitaln** and click **Submit**.

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5. You will be directed to the organization login screen.
6. Once you enter credentials of **Tommy** you will get redirected to the email OTP page as we have configured **MFA** for **Hospital n** Login Flow.
7. Try the steps from 1-7 with **Hospital a** details and users. You will only get one login step.

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